Curriculum Vitae

Uijoon Jeong

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RESEARCH INTERESTS

3D/4D Computer Vision, Dynamic Scene Understanding, Novel View Synthesis, Generative Models

EDUCATION

Sungkyunkwan University (SKKU), Seoul, South Korea Sep 2024 – Present

Master of Science in Immersive Media Engineering GPA: 4.26 / 4.5

Sungkyunkwan University (SKKU), Suwon, South Korea Mar 2014 – Aug 2024

Bachelor of Science in Semiconductor System Engineering

RESEARCH EXPERIENCE

Graduate Researcher, SKKU (Sungkyunkwan University), Sep 2024 - Present

Advisor: Prof. Eun-Seok Ryu

Dynamic 4D Object Segmentation

- Developed object-level segmentation for Ex4DGS by learning Gaussian-level semantic features from SAM2-generated pseudo-label masks.
- Constructed multi-view and temporal pseudo-labels to supervise object features in dynamic Gaussian representations.
- Enabled object-level extraction and rendering across viewpoints and time.

Novel-View Artifact Restoration in Sparse-View 3DGS

- Developed a confidence-guided diffusion framework to restore artifact-affected regions in novel views rendered by sparse-view 3DGS.
- Estimated spatial confidence maps using MEt3R-based view consistency between rendered novel views and input views.
- Applied confidence-aware flow-matching vector blending to spatially control diffusion refinement, encouraging stronger generation in low-confidence regions while preserving reliable content.
- Currently extending the framework with learned multi-view confidence using a VGGT encoder and DPT-based decoder, together with video diffusion for more consistent restoration.

PUBLICATIONS

Sparse-View 3D Gaussian Splatting Optimization By Using View Consistency-Based Confidence Maps and Diffusion Models
Uijoon Jeong, Yeonho Han, Eun-Seok Ryu
Journal of Broadcasting Engineering (JBE), 2026. (Accepted) [In Korean]

MagnetGS-Mesh: High-Quality Multi-Object Mesh Reconstruction via Adaptive Surface Optimization
Min-Su Park, Yeonho Han, **Uijoon Jeong**, Jun-Hyeong Park, Eun-Seok Ryu
ECCV (European Conference on Computer Vision), 2026

No-Reference Quality Assessment for Arbitrary-View Rendering with 3D Gaussian Splatting
Yeonho Han, **Uijoon Jeong**, Yu-Chan Cho, Eun-Seok Ryu
KIBME (The Korean Institute of Broadcast and Media Engineers Fall Conference), Nov. 2025. [In Korean]

Volumetric Video Object Extraction Based on 3D Gaussian Splatting Using 2D Segmentation Masks
Uijoon Jeong, Yeong-Gyu Kim, Eun-Seok Ryu
KIBME (The Korean Institute of Broadcast and Media Engineers Summer Conference), Jun. 2025. [In Korean]

*Note: Some publications list my name as Ui Joon Jeong or Ui-Joon Jeong.

STANDARDS

SKKU-ETRI Report on the JPEG RF Call for Datasets – 3D Gaussian Splatting Quality Assessment
Yeonho Han, **Uijoon Jeong**, Eun-Seok Ryu (SKKU), Gi-Mun Um, Hyon-Gon Choo (ETRI)
JPEG2026/M113021, 113rd JPEG Meeting of ISO/IEC JTC1/SC29/WG1, Oct.2026.

Object Segmentation Method in Volumetric Video Based on 3D Gaussian Splatting
Uijoon Jeong, Eun-Seok Ryu
Multi-dimensional Images Technology Standardization Forum, No. XDFK_01.0075/R0, Nov. 2025.

PATENTS

METHOD AND APPARATUS FOR EXTRACTING OBJECTS FROM MULTI-VIEW VIDEO SEQUENCES
Eun-Seok Ryu, **Uijoon Jeong**, Young-Gyu Kim, Reagan Koo
Research & Business Foundation of Sungkyunkwan University
No. 10-2025-0178487, Nov. 21, 2025.

AWARDS AND HONORS

Outstanding Research Award at the SKKU IME Annual Research Review 2025. Nov 2025
"*Volumetric Video Object Extraction Based on 3D Gaussian Splatting Using 2D Segmentation Masks*"

TEACHING EXPERIENCE

Teaching Assistant (T.A.), Immersive Video Processing, Department of Immersive Media Engineering, SKKU

Sep 2025 – Dec 2025

REFERENCES

Eun-Seok Ryu
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M.S. Advisor
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Kyungho Lee
Assistant Professor, Sungkyunkwan University
Email: kyungholee@skku.edu